Physics Writeup

Introduction

We are simulating light-matter interactions in an enhanced optical cavity in a λ-system set-up. The set-up is that of an optical cavity in which a three level system is placed inside an optical resonator, which one can imagine as a space between two mirrors. We assume the space within and surrounding the resonator has a background EM field with zero mean, but finite fluctuations. As such the surrounding space contains photon modes that can hold individual photons according to their defined wavevectors, $\vec{k}$. The λ-system has three possible states, $|g\rangle$, $|s\rangle$, and $|e\rangle$, known as the ground state, metastable state and excited state respectively. The states have energies in order $E_{|e\rangle} > E_{|s\rangle} > E_{|g\rangle}$. These sates can be easily implemented in Qutip by setting

$$|g\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |e\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |s\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{1}$$

```
# This is a Python code chunk
from qutip import basis
g = basis(3, 0)  # ground state
e = basis(3, 1)  # excited state
s = basis(3, 2)  # metastable state
```

The basis for the photons in the surrounding field is the Fock number basis $|n_k\rangle$, where n_k is the number of photons with wavevector of magnitude k, so for example, if the field has one photon with wavevector $\vec{k}_1$ and one with wavevector $\vec{k}_2$, then the field state is $|0...1...1....0\rangle$ with ones at position k_1 and k_2. Note, the importance of the mirrors in the resonator is so that photons continually bounce between them and stay within the resonator, allowing any one photon to interact with the matter system multiple times. This is what we mean by the system being cavity enhanced. We will now take the mirrors to be lossy such that photons can enter and escape the resonator. This will allow photons driven by an external laser to interact with the matter system in the simulation.

We describe the dynamics of the simulation as an electron in the matter system starting in the intermediate $|s\rangle$ state, then being pumped by laser drive to $|e\rangle$, and then finally decaying down to $|g\rangle$, and hence producing a photon that can either itsself interact with the system, or exit the resonator through the lossy mirrors. The transitions that occur are governed by the three possible transition processes in our system. 1. Stimulated Absorption - Matter system electron absorbs an incoming photon and is excited to a higher energy state. 2. Stimulated Emission - Incoming photon perturbs unstable, excited matter system electron, causing it to decay, producing an additional photon with the same wavevector as the incoming photon. 3. Spontaneous Emission - Unstable excited electron decays, producing a photon with wavevector in a random direction, not necessarily equal to that of the incoming photon. With this in mind, we define our Hamiltonian as

$$H = \Delta_c a_{\vec{k}}^\dagger a_{\vec{k}} + \Delta_l \sigma_{se}^\dagger \sigma_{se} + g(a_{\vec{k}}^\dagger \sigma_{ge} + a_{\vec{k}} \sigma_{ge}^\dagger) + \Omega(\sigma_{se} + \sigma_{se}^\dagger) \tag{2}$$

Where we have defined $\sigma_{se} = \mathbb{1} \otimes |s\rangle \langle e|$ as the system lowering operator from state $|e\rangle$ to $|s\rangle$ and accordingly $\sigma_{se}^\dagger = \mathbb{1} \otimes |e\rangle \langle s|$ as the raising operator from $|s\rangle$ to $|e\rangle$. Similarly, $\sigma_{ge} = \mathbb{1} \otimes |g\rangle \langle e|$ is the lowering operator from $|e\rangle$ to $|g\rangle$ and $\sigma_{ge}^\dagger = \mathbb{1} \otimes |e\rangle \langle g|$ is the raising operator from $|g\rangle$ to $|e\rangle$. Furthermore, we define $a_{\vec{k}} \equiv a_{\vec{k}} \otimes \mathbb{1}$ as the field annihilation operator that destroys a photon of wave vector $\vec{k}$, and $a_{\vec{k}}^\dagger \equiv a_{\vec{k}}^\dagger \otimes \mathbb{1}$ create a photon at said $\vec{k}$. This can be implemented in Qutip by

```
# This is a Python code chunk
from qutip import *
Ncav=4,
detune_c=1.0,
detune_l=1.0,
g_val=1.0,
rabi=2.0,
Ncav = Ncav  # photon truncation
# Operators on atom
sigma_ge = g * e.dag()
sigma_se = s * e.dag()
I_atom = qeye(3)

# Operators on cavity
a = destroy(Ncav)
I_cav = qeye(Ncav)

# Tensor operators
a = tensor(a, I_atom)
sigma_ge = tensor(I_cav, sigma_ge)
sigma_se = tensor(I_cav, sigma_se)
# Parameters
```

```
detune_c = detune_c  # Detuning between cavity resonance and atomic |g |e transition freque
detune_l = detune_l  # Detuning between classical laser and atomic |s |e transition frequen
g_val = g_val  # Atom-cavity coupling strength (vacuum Rabi frequency) for |g |e transition
rabi = rabi  # Rabi frequency of classical laser driving |s |e transition

# Hamiltonian
H = (
    detune_c * a.dag() * a
    + detune_l * sigma_se.dag() * sigma_se
    + g_val * (a.dag() * sigma_ge + a * sigma_ge.dag())
    + rabi * (sigma_se + sigma_se.dag())
)
```

$\Delta c = |\omega_L - \omega_{ge}|$ is the detuning ie frequency difference between laser drive and the frequency of $E_{|e\rangle} - E_{|g\rangle}$ energy difference. Similarly, $\Delta_l = |\omega_L - \omega_{se}|$ is the detuning between laser drive frequency and frequency of $E_{|e\rangle} - E_{|s\rangle}$ energy difference. g is the coupling strength coefficient between the matter system and the field, and Ω is the Rabi oscillation frequency representing lasing amplitude.

Notice that since the mirrors are lossy, not only can laser driven photons enter the resonator, but photons from within the resonator can also leak out and souple to modes outside the resonator. We take there to be a continuum of modes outside, as supposed to modes within the resonator which are restricted to half-integer wavelengths fitting within resonator length. As such, the contiuum of modes external to the resonator can be taken as an environment, meaning that the set-up is effectively that of an open quantum system. As such, the state exists as a pure state density matrix $\rho = \rho_{sys} \otimes \rho_{env}$. Since we are only interested in system dynamics, we can trace over environement degrees of freedom and work with just ρ_{sys}. Note, by system here we mean the matter system and field modes within the resonator, not just the matter system itself. The structure of ρ_{sys} is as

$$\rho_{sys} = \begin{pmatrix} \rho_{ee} & \rho_{es} & \rho_{eg} \\ \rho_{se} & \rho_{ss} & \rho_{sg} \\ \rho_{ge} & \rho_{gs} & \rho_{gg} \end{pmatrix} \tag{3}$$

Here ρ_{xx} is the probability (ie population) of the electron being in state $|x\rangle$ and must satisfy $\Sigma_j \rho_{jj} = 1$, while ρ_{xy} is the coherence between the $|x\rangle$ and $|y\rangle$ states that satisfies $\rho_{xy} = \rho^*_{yx}$. The state is initialized in $\rho_{sys} = |0000\rangle \langle s|$, ie with resonator of four total cavity modes of zero photons and matter system in $|s\rangle$. This is done in Qutip by

```
init_cav=0,
init_sys=2,
psi = tensor(basis(Ncav, init_cav), basis(3, init_sys))
```

Due to the open quantum nature, the dynamics can be modeled using the Lindblad Master Equation given by

$$\frac{d\rho_{\text{sys}}}{dt} = -i[H, \rho_{\text{sys}}] + \left(L_1 \rho_{\text{sys}} L_1^\dagger - \frac{1}{2}\{L_1^\dagger L_1, \rho_{\text{sys}}\}\right) + \left(L_2 \rho_{\text{sys}} L_2^\dagger - \frac{1}{2}\{L_2^\dagger L_2, \rho_{\text{sys}}\}\right),$$
$$L_1 = \sqrt{\kappa}\, a_{\vec{k}},$$
$$L_2 = \sqrt{\gamma}\, \sigma_{ge}$$

Here L_1 and L_2 are the collapse operators that represent losses from the system to the environment. κ here is the cavity photon decay rate of photons leaking through the mirrors, while γ is the spontaneous decay rate from $|e\rangle$ to $|g\rangle$. These are the two processes by which energy leaves the resonator (system). This equation can be implemented into Qutip using MESolve, where the collapse operators are taken as arguments

```
from qutip import *
kappa=0.2,
gamma=0.05
c_ops = [np.sqrt(kappa) * a, np.sqrt(gamma) * sigma_ge]
psi = tensor(basis(Ncav, init_cav), basis(3, init_sys))

result = mesolve(
    H, psi, times, c_ops, [a.dag() * a, sigma_ge.dag() * sigma_ge, P_g, P_s]
)
```

Note, the third argument in MESolve is the list of operators whose expectation values are to be returned. Finally, we can define the cooperativity to be

$$C = \frac{g^2}{\kappa\gamma} \tag{4}$$

This is a measure of the degree of cavity enhancement ie the number of times any given photon interacts with the matter system before leaking out of the resonator

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